MIL-E-1/1494B(NAVY)
15 August 1966
SUPERSEDING
MIL-E-1/1494A (NAVY)
1 MARCH 1965
(See Note 19)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, PULSE AMPLIFIER

TYPES 8428 AND 8455
a b

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Grid Controlled, Electron Multiplier

DIMENSIONS: See figure 1.

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Test code	Ef	Eb	Edy1	Edy2	Edy3
Unit:		V	Vdc Note 1	Vdc Note 1	Vdc Note 1	Vdc Note 1
Maximum:	a	6.6	1000	-1900	-1400	-900
	b	6.6	1600	-1900	-1400	-900
Minimum:	a,b	6.0	-	0	0	0
TEST CONDITIONS:	a	6.3	800	-1500	-1100	-700
	b	6.3	1300	-1500	-1100	-700

ABSOLUTE-MAXIMUM RATINGS:

Parameter:		Edy4	Edy5	Po	po	eo	Pp
Unit:		Vdc Note 1	Vdc Note 1	W Note 2	kw Note 2	v	W Note 3
Maximum:	a	-400	-	30	3	700	12
	b	-400	-	44	5	1200	12
Minimum:	a,b	0	-	-	-	-	-
TEST CONDITIONS:	a	-300	-	-	-	-	-
	b	-300	-	-	-	-	-

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ABSOLUTE-MAXIMUM RATINGS:

		Test code					
Parameter:		Ec3	Ec2	Ec1	Egk	egk	Ek
Unit:		Vdc Note 1	Vdc Note 1	Vdc Note 1	Vdc	v Note 5	Vdc Note 1
Maximum:	a,b	-2100	-2100	-2450	-50	40	-2400
Minimum:	a,b	0	0	0	-10	---	0
TEST CONDITIONS:	a,b	-1600	-1600	---	-15	+5.0	-1900

ABSOLUTE-MAXIMUM RATINGS:

Parameter:		tp	TA	Du
Unit:		us	°C	% Note 4
Maximum:	a,b	100K	+80	2.0
Minimum:	a,b	---	-55	---
TEST CONDITIONS:	a,b	10	---	1.0

BASE CONNECTIONS:

Pin No.		1	2	3	4	5	6	7	8	9	10	11	12	13	Coax. Conn.
Element	a,b	int. con.	int. con.	A	Dy4	Dy3	Dy2	Dy1	G3	G2	int. con.	h	h k	G1	Dy5

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required (Note 12)

5 Preparation for delivery

E-50.2 Holding period

E-50.3 Preheating

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METHOD OR PARA.	REQUIREMENT OR TEST	Test code	CONDITIONS	AQL (% Defect.)	Insp. Level or Code	Sym	Limits		Units
							Min	Max	
	Qualification								
1006	Salt spray	a,b	Note 6	-	-	-	-	-	-
--	Shock	a,b	No voltages applies; note 7						
--	Post shock end point	a	Operation (1)	-	-	eo	600	-	V
		b	Operation (1)	-	-	eo	1000	-	V
1031	Variable-frequency Vibration (2)	a,b	Operation (1); t=2 hours in each of 2 planes X and Z; Note 6	-	-	-	-	-	-
1031	Variable-frequency Vibration (3)	a,b	Operation (1); t=2 hours in each of 2 planes X and Z; Note 15	-	-	-	-	-	-
--	Post vibration (2) and (3) End Point	a,b	Operation (1)	-	-	Δeo t	-	10	%
--	Operation (3)	a,b	Operation (1) except RL=250 ohms; Ebb=400Vdc; Notes 8, 14 and 16	-	-	eo tre tad Δeo tp Δeo DU Δeo egk	325 - - - - -	- 10 20 1.5 1.0 1.0	V ns ns % % %
--	Operation (4)	a,b	Operation (1) except RL=250 ohms; Ebb=600 Vdc; Notes 8, 14 and 16	-	-	eo tre tad Δeo tp Δeo DU Δeo egk	475 - - - - -	- 10 20 1.5 1.0 1.0	V ns ns % % %

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METHOD OR PARA.	REQUIREMENT OR TEST	Test code	CONDITIONS	AQL (% Defect.)	Insp. Level or Code	Sym	Limits Min	Limits Max	Units
--	Operation (5)	a,b	Operation (1) except RL=250 ohms; Ebb=800 Vdc; Notes 8, 14 and 16	-	-	eo	675	-	v
						trc	-	10	ns
						tad	-	20	ns
						Δeo tp	-	1.5	%
						Δeo DU	-	1.0	%
						Δeo egk	-	1.0	%
	Quality Conformance Inspection, part 1 (Note 9)								
--	Operation (1)	a	RL=150 ohms; CL=15pf; notes 10 and 16	-	-	eo	600	-	v
		b	RL=250 ohms; CL=15pf; notes 10 and 16	-	-	eo	1000	-	v
--	Output pulse rise time	a,b	Operation (1)	-	-	trc	-	10	ns
--	Input to output delay time	a,b	Operation (1); Note 11	-	-	tad	-	20	ns
--	Plate current	a	Operation (1) except Ebb/eo=600 v	-	-	Ib	-	65	mAdc
--	Plate current	b	Operation (1) except Ebb/eo=1000 v	-	-	Ib	-	68	mAdc
--	Interpulse Plate current	a,b	Operation (1) except egk=-15v	-	-	Ib	-	5	mAdc
	Quality Conformance Inspection, Part 2								
--	Output pulse droop	a,b	Operation (1); Note 13	6.5	L6	Δeo tp	-	1.5	%

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METHOD OR PARA.	REQUIREMENT OR TEST	Test Code	CONDITIONS	AQL % Defect.	Insp. Level or Code	Sym	Limits		Units
							Min	Max	
1256	Positive grid current	a,b	Operation (1)	6.5	L6	Ic	-	0.1	mAdc
1301	Heater current	a,b		6.5	L6	If	1.1	1.3	A
1331	Direct interelectrode capacitance	a,b	No shield; g1 to all except internal connections and metal envelope grounded			Cin(1)	-	13	pf
			No shield; g1 to all including internal connections and metal envelope			Cin(2)	-	17	pf
			No shield; dy5 to all except internal connections and metal envelope grounded	6.5	L6	Cout(1)	-	70	pf
			No shield; dy5 to all including internal connections and metal envelope			Cout(2)	-	100	pf
1031	Variable-frequency vibration (1)	a,b	Ef=6.3V; no other voltages t=20 minutes in each of 2 planes X and Z	6.5	L6	-	-	-	-
--	Post vibration (1) End Point	a	Operation (1)	-	-	eo	600	-	v
		b	Operation (1)	-	-	eo	1000	-	v
--	Operation (2)	a	Ebb=1000Vdc max; RL=150 ohms CL=15pf; Note 10	6.5	L6	eo	700	-	v
		b	Ebb=1600Vdc Max; RL=250 ohms CL=15pf; Note 10	6.5	L6	eo	1200	-	v
--	Operation (6)	b	Operation (1) except Du=2% Ebb/eo=600v	6.5	L6	Ebb	-	750	Vdc
--	Dynode current	a	Operation (1); Note 18	6.5	L6	Idy	-	22	mAdc
--	Dynode current	b	Operation (1); Note 18	6.5	L6	Idy	-	25	mAdc

MIL-E-1 1-3-1 (AV)

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METHOD OR PARA.	REQUIREMENT OR TEST	Test Code	CONDITIONS	Sym	Limits		Units
					Min	Max	
	Quality Conformance Inspection, Part 3						
4.7	Life test	a	Group D; t=500 hr. (min.) RL=150 ohms; CL=15 pf; Ebb/eo=600 v; Note 17				
		b	Group D; t=500 hr. (min.) RL=250 ohms; CL=15 pf; Ebb/eo=1000 v; Note 17				
4.7.3	Life test end point	a	Operation (1) except Ebb/eo=600 v	Ebb	-	900	Vdc
		b	Operation (1) except Ebb/eo=1000 v	Ebb	-	1450	Vdc

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Note 1: Voltages given with respect to ground. Dynode 5 is normally connected to ground through the load impedance. The following are the maximum voltage differences permitted between elements.

(a) 500 volts between cathode and dynode 1; dynodes 1 and 2; dynodes 2 and 3; and dynodes 3 and 4.
(b) 400 volts between dynodes 4 and 5.
(c) 300 volts between grids 2, 3, and cathode.
(d) 100 volts between grid 1 and cathode.

Note 2: Output taken from dynode 5.

Note 3: Plate dissipation is defined as:

$P_p=(E_{bb}-e_o)I_b$. Where output is taken from dynode 5.

Note 4. For duty cycles greater than one percent, see figure 4. If duty cycle exceeds one percent, other maximum ratings shall not be exceeded.

Note 5: The maximum positive grid voltage with respect to the cathode shall not exceed 40 volts peak.

Note 6: This test shall be performed on two tubes.

Note 7: The tube shall be subjected to six impacts in each of two positions, X and Z, each impact to be 50 g at 11 ± 1 millisecond. This test shall be performed on three tubes.

Note 8: $\frac{\Delta e_p}{e_{gk}}$ is change in e_o due to change in e_{gk} from e_{gk} = + 1.0 v to e_{gk} -.0 v.

Note 9: These tests shall be performed on a 100 percent basis.

Note 10: The tube shall be operated as a pulse amplifier in the test circuit shown on figure 2. Input pulse rise time shall be 7 ± 2 ns (from 10 percent to 90 percent of pulse amplitude). Measure e_o at 0.5 tp.

Note 11: Input to output delay time is defined as the time interval between the point on the rising portion of the grid pulse which is 50 percent of the maximum pulse amplitude and the point on the rising portion of the output pulse which is 50 percent of the maximum pulse amplitude.

Note 12: The activity responsible for the qualified products list is the Naval Ship Engineering Center, Department of the Navy, Washington, D. C., 20360, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

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Note 13: See figure 3 for definition of pulse droop.

Note 14: The test shall be performed at Du=0.01 percent, Du=0.1 percent and Du=1.0 percent.

Note 15: Two additional tubes shall be vibrated in each of two positions through the frequency ranges at the amplitudes and for the following duration:

Frequency Range cps	Duration Minutes	Double Displacement Inch
5-15-5	40	0.18
16-25-16	40	0.20
26-33-26	40	0.10

The time for gradually covering the frequency range shall be 1 to 5 minutes.

Note 16: There shall be no noise or oscillation, on the output pulses, in excess of 5 volts peak-to-peak, excluding pulse ringing or power supply hum.

Note 17: The tube shall be operated as a pulse amplifier in the test circuit shown on figure 2. Input pulse rise time shall be 100 ns maximum (from 10 percent to 90 percent of pulse amplitude). Measure eo at 0.5 tp.

Note 18: The total current of dy1, dy2, dy3, and dy4 shall be within the limit specified herein.

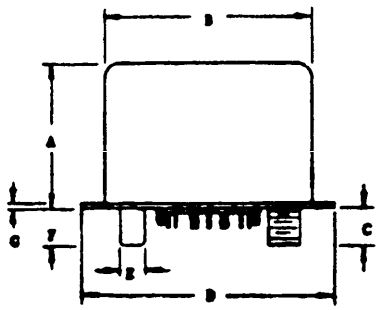
Note 19: CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Review activity:
Navy - SH

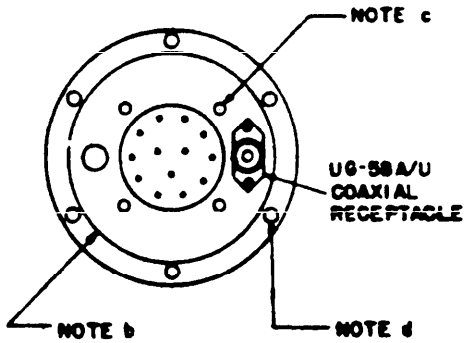
User activities:
Navy - AS, OS, CG, MC

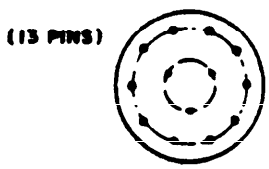
Preparing activity:
Navy - SH
(Project 5960-N716(NAVY))

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DIM.	AQL (PERCENT DEFECTIVE)	INSP. LEVEL	LIMITS MIN.	LIMITS MAX.
Quality conformance inspection, part 2				
A	6.5	L6	3.190	3.350
B	6.5	L6	4.650	4.850
C	6.5	L6	——	0.750
D	6.5	L6	5.700	5.800
GENERAL DIMENSIONS				
E	0.520			
F	0.700			
G	0.125			





NOTES

a. All dimensions in inches except as noted.

b. 4.750 inch diameter chassis clearance hole required.

c. Four 6-32 x 0.225 inch long studs equally spaced on 3.140 inch diameter.

d. Six 0.173 inch diameter holes equally spaced on 5.375 inch diameter.

e. Pin alignment shall be inspected in accordance with Appendix D, para. 30.3.4.2 using gage shown on figure 5.

SH 8442

Figure 1 - Outline drawing of tube types 8428 and 8455.

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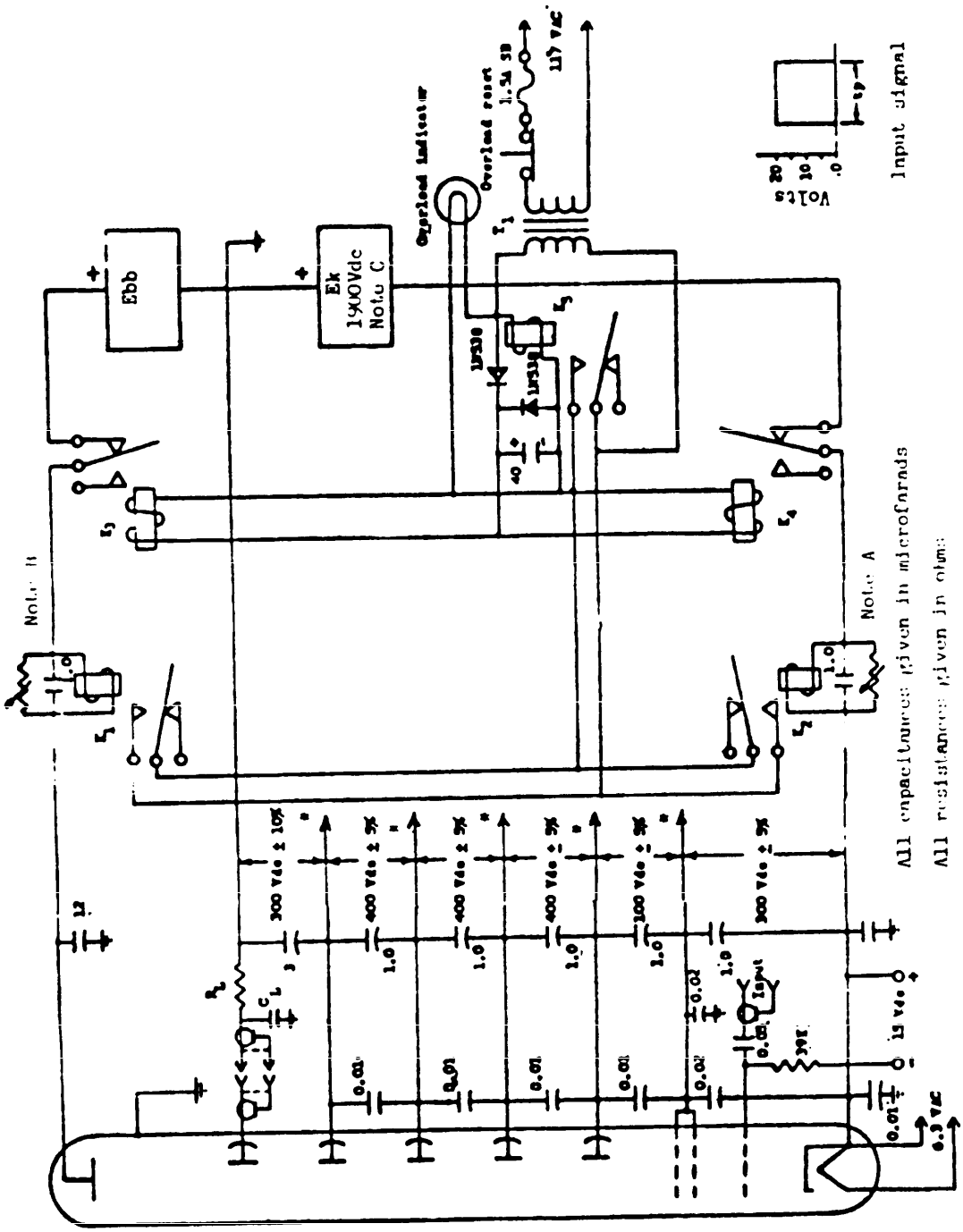


Figure 2 - Test circuit.

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NOTES:

A. The variable resistor across relay K_2 is adjusted such that the relay is activated at approximately 45 ma through the resistor-relay coil combination.

B. The variable resistor across relay K_1 is adjusted such that the relay is activated at approximately 80 ma through the resistor-relay coil combination.

C. Cathode voltage (Ek) shall be regulated to ± 1 percent with a maximum ripple of 5 volts peak-to-peak.

D. Voltage supply shall be capable of maintaining these electrode voltages within the tolerances specified herein.

Parts list:

K_1, K_2 - Automatic electric k 2012-1644 (modified with 1/16 inch teflon sheet between contacts and coil case), or equivalent.

K_3, K_4 - Advance relay HV/12, or equivalent

K_5 - Line electric MKP1A - 115Vac, or equivalent

T_1 - Stancor P6161 (isolation transformer), or equivalent

Figure 2 - Test circuit (Cont d).

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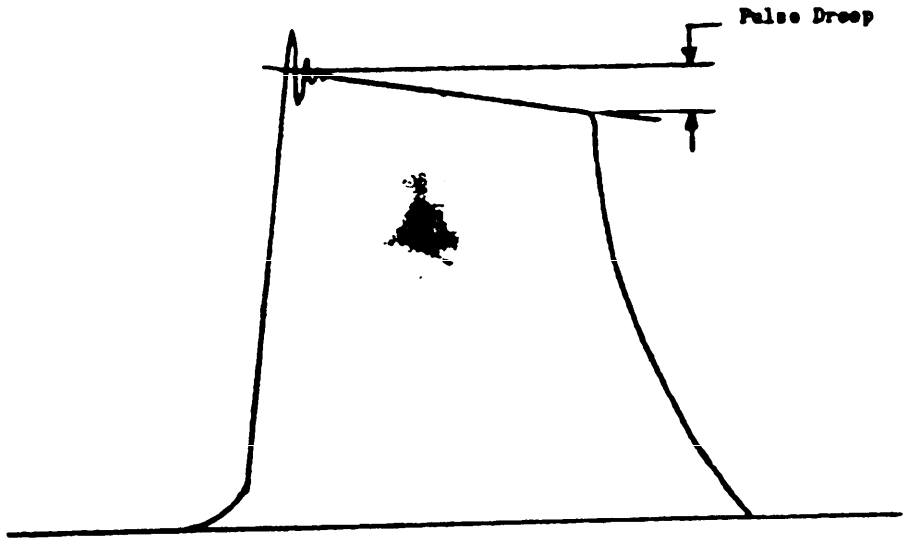


Figure 3 - Pulse Droop Definition

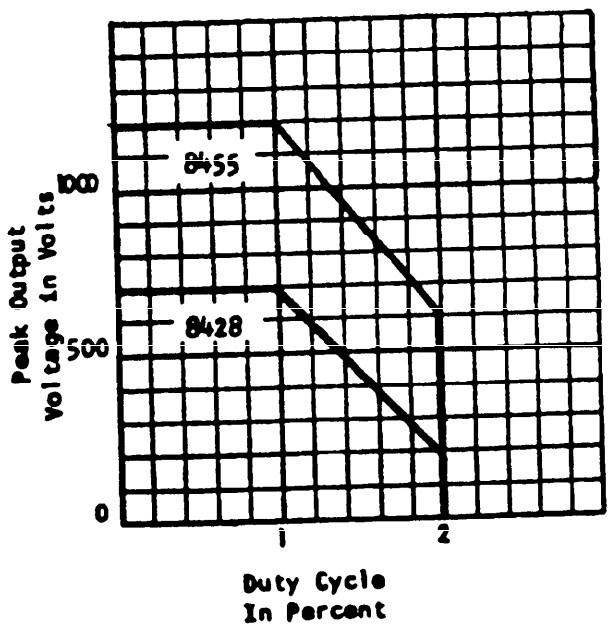
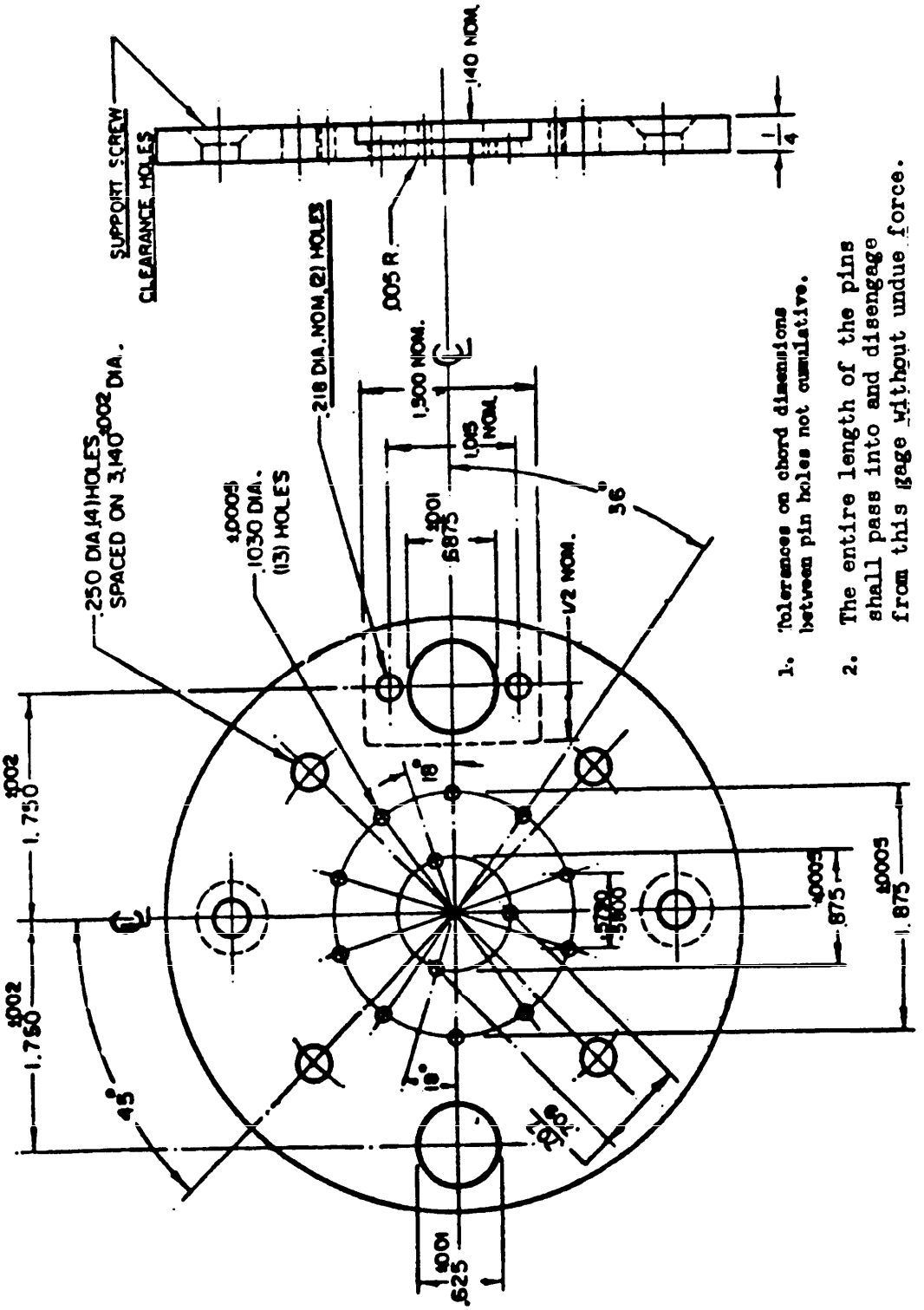


Figure 4 - Maximum rating curve for output voltage versus duty cycle.

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1. Tolerances on chord dimensions between pin holes not cumulative.
2. The entire length of the pins shall pass into and disengage from this gage without undue force.

Figure 5 - Base-pin alinement gage.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT -- ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-86384

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